FRANCISCO MORENO DE LA CRUZ

Backend Developer

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Profile

Backend Developer with a Higher Vocational Diploma in Multiplatform Application Development and knowledge of C, C++, Python, SQL, and Linux. Additional hands-on training at 42 Madrid – Fundación Telefónica, involving projects in systems programming, networking, and algorithms. Keen to continue growing professionally in backend development and software architecture.

Professional Experience

Backend Developer - Mirelia Digital Talent
03/2024 – 08/2024

TypeScript • Prisma ORM • PostgreSQL • Docker • Git

Madrid, Spain

- Developed the backend using TypeScript, Prisma ORM, and PostgreSQL.
- Designed and implemented REST APIs, endpoints, business logic, and data access.
- Designed database models and relationships using Prisma ORM.
- Configured the development environment with Docker and managed code and version control using Git.

Freelance Developer
06/2025 – Present

Python • C++ • SQL • Git

- Developed an IRC chat server in C++, implementing socket-based client-server communication and managing multiple connections, channels, and commands.
- Developed a GPS application in Python, handling data processing, route logic, and geographic information.
- Implemented a Minesweeper-solving algorithm in Python using board analysis, logical inference, and probability calculations.
- Developed an Excel-style administrative application in Python, connected to an SQL database for querying, editing, and managing structured information.

Education

Software Developer
11/2024 – 06/2026

42 Madrid – Fundación Telefónica

• C, C++, Linux, Python, Shell, SQL, Docker, Git, TCP/IP, Algorithms & Data Structures

Higher Vocational Diploma in Multiplatform Application Development
09/2022 – 07/2024

CampusFP

• Python, SQL, Java, PHP, Git, Algorithms & Data Structures

Computer Engineering
09/2019 – 06/2022

University of Alcalá de Henares

• C, C++, SQL, Python, Algorithms & Data Structures, Operating Systems, Statistics

Higher Vocational Diploma in Industrial Automation and Robotics
09/2017 – 07/2019

IES Luis de Lucena

Skills

Programming Languages: C • C++ • Python • SQL • Shell

Systems & DevOps: Linux • Git • Docker

Software Engineering: Algorithms & Data Structures • Object-Oriented Programming • Software Architecture • Networking

Soft Skills: Self-directed Learning • Problem-solving • Teamwork • Critical Thinking

Selected Projects

Minishell – C

LinkedIn | GitHub

Developed a Bash-inspired shell supporting command parsing, environment variables, and built-in commands. Implemented pipes, redirections, process management, signals, and exit codes.

Minesweeper AI – Python

LinkedIn | GitHub | PyAutoGui

Developed an algorithm to solve Minesweeper games through pixel analysis, logical inference, and probability calculations. Implemented board analysis, decision-making, and safe-move selection.

Languages

Spanish	Native
English	B2